

Food and Health (Part - 3)

Some of the major malnutrition diseases in the world are:

Name of the disease	Occurs due to deficiency of
Beriberi	Vitamin
Kwashiorkor	Proteins
Scurvy	Vitamin C
Pellagra	Vitamin
Rickets	Vitamin D, calcium or phosphorus

Symptoms	Foods that contain the nutrient
Swollen abdomen and shortness of breath	Wheat gram, nuts and most of the vegetables.
Swollen abdomen and diarrhea	This has to be carefully treated. You can't give protein rich diet straightly
Loss of hair and teeth swelling and pain in joints	Citrus fruits, guavas, tomatoes.
Diarrhea, insomnia and memory loss	Whole grains peanuts and lentils.
Skeletal deformation, pain in bones	Fatty fish, cow's milk egg yolk.

Hunger is something which our body makes us feel when it needs food. We want to eat something when we feel hungry and our body needs energy to do work. People fall prey to the above mentioned diseases when they do not get food and some of them even die due to hunger. To fight from situations like this, the food stored in god owns

is used. A godown is like a storehouse where the extra grains and cereals are preserved so that they can be used when there is not sufficient production of food. But a proper care is needed as sometimes a lot of food is wasted due to improper storage of food.



Fig: A Godown (used for storage of grains and cereals)

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How do we taste food? When we taste food we get different tastes from different foods. Our tongue helps us to taste the food with the help of our brain. There are different taste buds in our tongue which can detect different tastes. One region of our tongue can taste sour while the other can taste sweet and so on.

Digestion starts from mouth: The role of mouth is not only to take the food inside and chew it but also to digest it. The salivary glands present in mouth start producing saliva when the smell of food starts coming. They produce more saliva when the food is taken inside the mouth. There is also secretion of juices which help the food to break down further. So in this way the saliva starts the digestion of food from mouth. Sometimes when we eat rice or chapatti, etc., we get a taste of sweetness after chewing them for some time. This is because rice or the chapatti contains starch which is converted to sugars with the action of saliva.

Our body changes the food to glucose which then gets mixed with the bloodstream.

What is glucose?

It is an important energy source that is needed by all the organs of our body. It comes from what we eat.

Glucose is given to the person who is ill when his body is not in the state to form it. It is given to the patient so as to provide energy to his body.



Fig: A Glucose drip

Plants also need food: Like any other living being, plants also need food. You have already studied in previous classes that plants need water, air and sunlight to prepare their food. This process is known as photosynthesis.

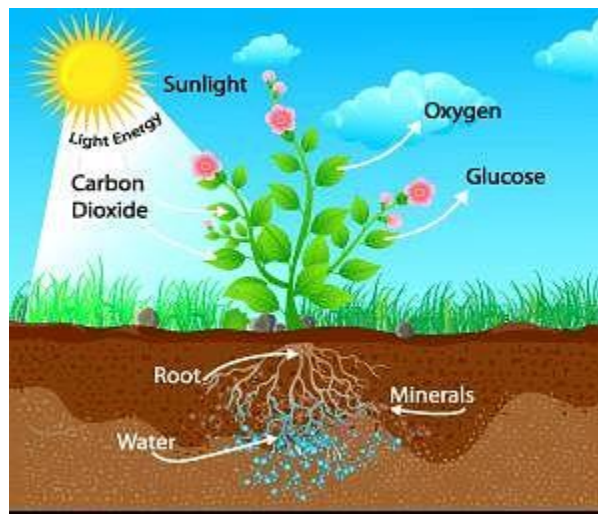


Fig: Photosynthesis

Some of the plants like pitcher plant, Venus fly trap, sundew etc., eat insects and hence are called insectivorous.

Food for animals: All animals need food to survive. There are different types of food that different animals eat.

The types of animals depending on their food are:

1. **Herbivores:** These are the animals which eat plants only. For example: cow, goat and horse, etc.
2. **Carnivores:** These animals eat other animals such as lion, leopard and eagle, etc.
3. **Omnivores:** These animals eat both plants and animals. For example: human beings, bears and rats.

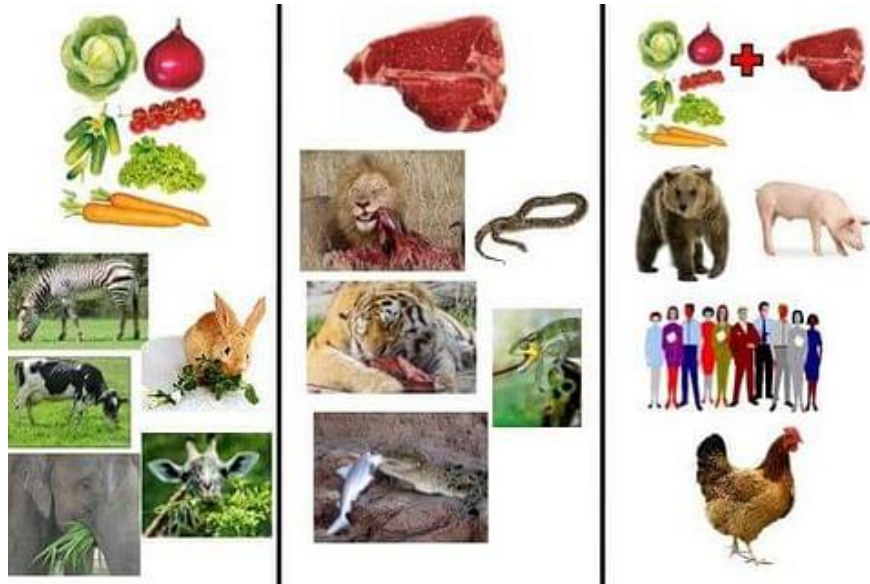


Fig: (a) Herbivorous (b) Carnivorous (c) Omnivorous

So now you can see, that all the animals have different types of food. Sometimes one animal eats the other and so on. Together they form a chain called food chain.

For example:

1. Earthworms are eaten by birds which in turn are eaten by snake.
2. Leaves are eaten by giraffes which are eaten by lion.

Many food chains put together is called food web.

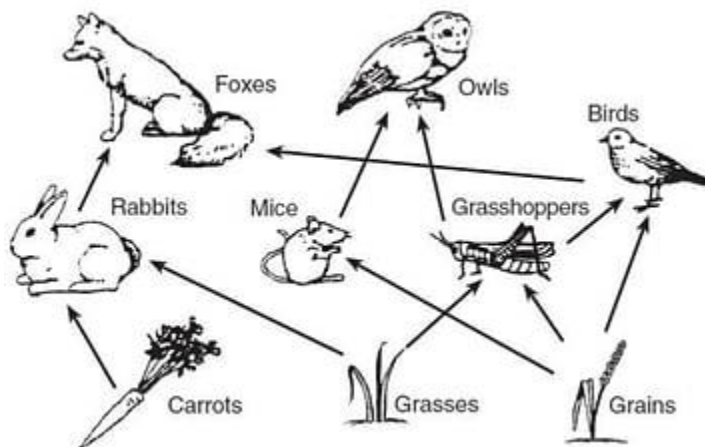


Fig: A food web

As we can see that growing food is not easy and many people do not get to eat proper amount of food, we should try to minimize the wastage of food. We can do so in these ways:

- We should buy only those things which we are going to use and those too in required quantity.
- We should keep the eatables properly so that they do not get spoiled.
- We should cook the food according to the need.

By following these steps we can minimize the wastage of food and we can fight against situation like famine.